Practical Numerical Simulations HW1

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1 Fourth-Order Runge–Kutta Method

We first consider the scalar ODE

$$\frac{dx}{dt} = (t-1)^2\,(x+1), \quad x(0) = 0 \tag{1}$$

and use the Runge-Kutta method with progressively halved step sizes h. This problem serves to confirm fourth-order convergence and to illustrate how the global error decreases as h is refined. First we solve this equation analytically,

$$\int_{x(0)}^{x(t)} \frac{dx}{x+1} = \int_0^t (t^2 - 2t + 1)dt \tag{2}$$

$$\Big[\ln(x+1)\Big]_{x(0)}^{x(t)} = \left[\frac{t^3}{3} - t^2 + t\right]_0^t \tag{3}$$

$$\ln(x(t)+1) = \frac{t^3}{3} - t^2 + t \tag{4}$$

Taking the exponential of both sides and rearranging we get our analytical solution,

$$x(t) = \exp\left(\frac{t^3}{3} - t^2 + 1\right) - 1 \tag{5}$$

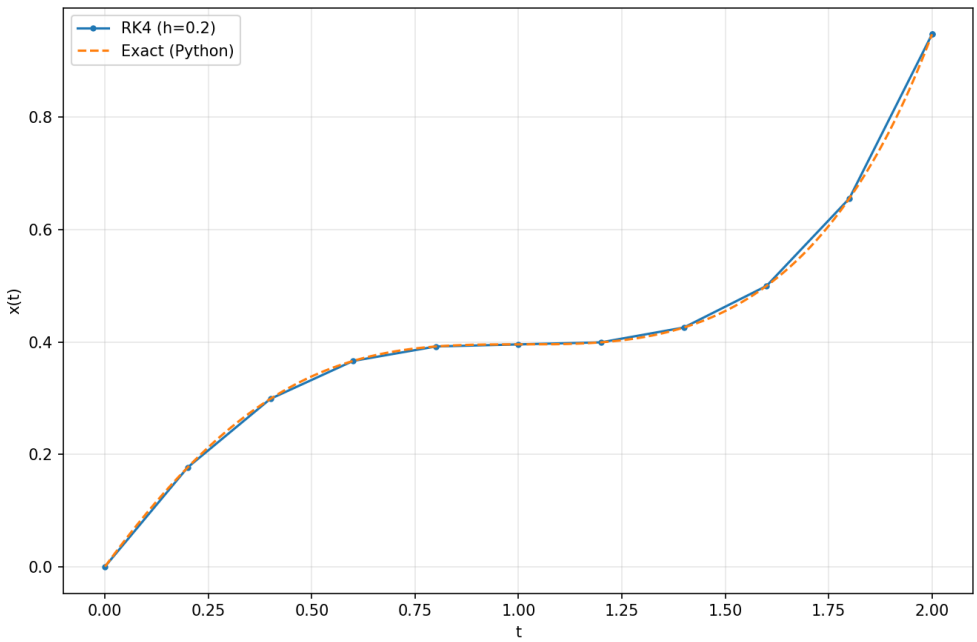

Figure 1: Analytical vs Approximate Solution on $t \in [0, 2]$ for $h = 0.2$

The approximate solution for $x(2)$ using a step size of $h = 0.2$ correct to 8 significant figures is 0.94772125. The data points used to make this plot are in `rk4_solution.csv` in Appendix B.

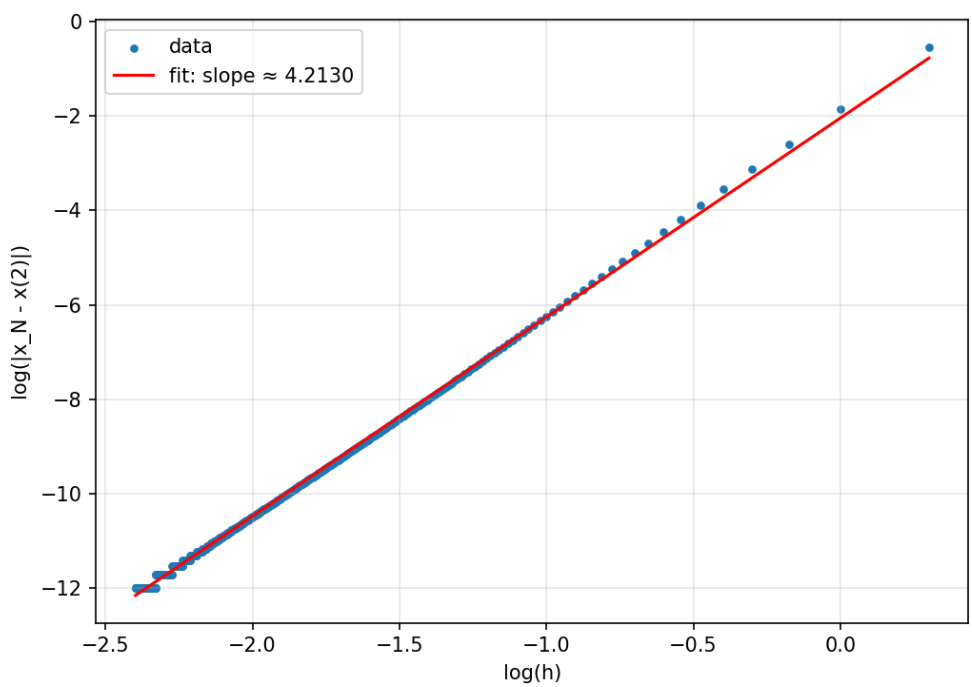

Figure 2: $\log_{10}(|x_N - x(2)|)$ vs. $\log_{10}(h)$, log-log plot of the Runge-Kutta global error against the step size

$$\log_{10}(|x_N - x(2)|) \approx (4.2130)\log_{10}(h) + c \tag{6}$$

$$|x_N - x(2)| \approx (10^c)(h^{4.2130}) \tag{7}$$

The global error $|x_N(2) - x(2)|$ was measured for many step sizes h and was plotted on a log-log scale. A straight line was fitted to this data which gives a slope of $m = 4.2130$, which can then be used illustrate the fourth order accuracy of the Runge-Kutta method.

$$\text{error}(h) \propto h^4 \tag{8}$$

To find a solution of (1) accurate to 8 significant figures, a .csv file (see `rk4_N_table.csv` in Appendix B) was made containing a range of step sizes h for an increasing number of iterations N evaluating the solution at $x(2)$ and comparing to the exact solution at $t = 2$. Reading off the .csv file, the approximated solution is accurate to 8 significant figures when $N = 87$ which corresponds to a step size of,

$$h = \frac{t_f}{N} = \frac{2}{87} = 0.022988506 \tag{9}$$

leading to an absolute error of 1.018×10^{-9}.

2 Runge–Kutta for a Fourth-Order ODE

We now solve the fourth-order ODE:

$$10\,\frac{d^4y}{dx^4} + 50\,\frac{d^2y}{dx^2} + x\,y^3 = 0, \tag{10}$$

with initial data

$$y(0) = 1, \qquad \frac{dy}{dx}(0) = \frac{d^2y}{dx^2}(0) = \frac{d^3y}{dx^3}(0) = 0. \tag{11}$$

First we reduce this to a first order system by introducing the state vector $\mathbf{u} = (u_1, u_2, u_3, u_4)^\top = (y, y', y'', y^{(3)})^\top$. Then the problem is equivalent to the first–order system:

$$\mathbf{u}'(x) = \mathbf{F}\big(x, \mathbf{u}(x)\big), \qquad \mathbf{u}(0) = (1, 0, 0, 0)^\top, \tag{12}$$

with

$$\mathbf{F}(x, \mathbf{u}) = \begin{pmatrix} u_2 \\ u_3 \\ u_4 \\ -5\,u_3 - 0.1\,x\,u_1^3 \end{pmatrix}. \tag{13}$$

We go from $x = 0$ to $x = 30$ using the Runge–Kutta method with step size h. At each grid point x_n with state $\mathbf{u}_n$, the Runge-Kutta method forms the stage vectors $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4$ via evaluations of $\mathbf{F}$, and updates $\mathbf{u}_{n+1}$ by the standard Runge-Kutta formula. Full stage definitions and the update are listed in the Appendix.

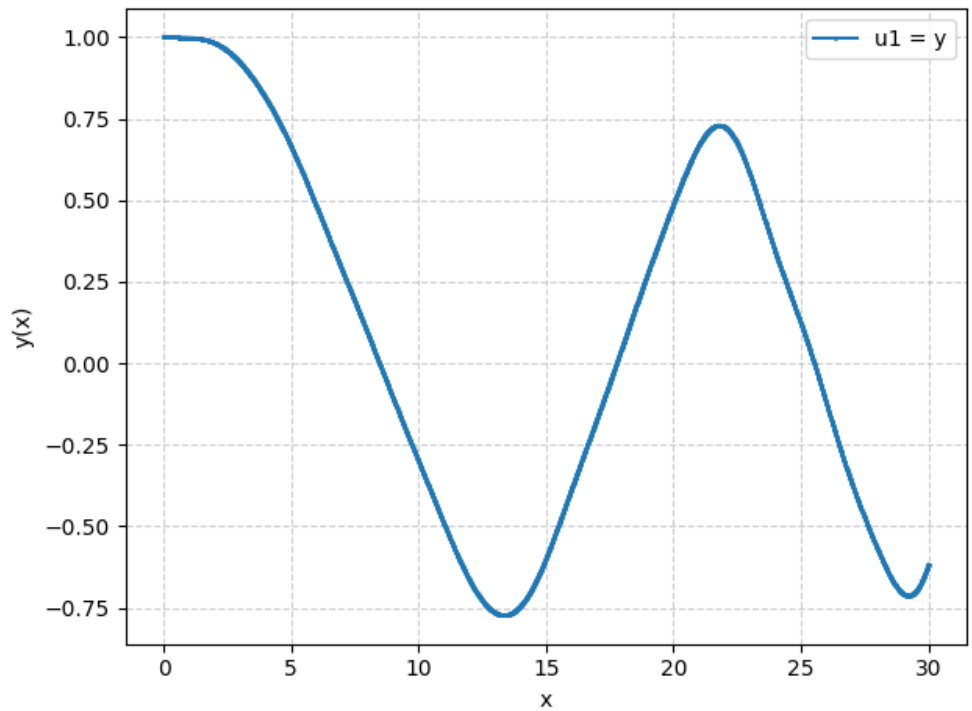

Figure 3: Computed $y(x)$ vs. x using higher-order Runge-Kutta method for $h = 0.0125$

The data points used to make this plot are in `pns3_solution.csv` in Appendix B. The approximated value for a step size of $h = 0.0125$ was found to be:

$$y(30) \approx -0.61703944.$$

The change between subsequent step sizes was taken to be less than 5×10^{-7},

$$\left| y_{\frac{h}{2}}(30) - y_h(30) \right| < 5 \times 10^{-7}, \tag{14}$$

which corresponds to accuracy of six significant figures. The convergence history is summarized in Table 1 below.

Step Size (h)	**Approximate** $y_h(30)$	**Relative Error from previous** $y_h(30)$
0.4	-0.61773259	6.983×10^{-3}
0.2	-0.61712891	6.037×10^{-4}
0.1	-0.61704598	8.293×10^{-5}
0.05	-0.61703987	6.110×10^{-6}
0.025	-0.61703946	4.1×10^{-7}
0.0125	-0.61703944	2×10^{-8}

Table 1: Runge-Kutta refinement for $y(30)$. The relative change is equation (14)

Since the threshold in (14) is first satisfied at $h = 0.025$, that certifies that $y(30)$ is accurate to six significant figures for this step size.

A Additional Figures

2. $10\frac{d^4y}{dx^4} + 50\frac{d^2y}{dx^2} + xy^3 = 0 \qquad y(0) = 1\ ,\ y'(0) = y''(0) = y'''(0) = 0$

$u_1 = y\ ,\ u_2 = y'\ ,\ u_3 = y''\ ,\ u_4 = y'''$

$u_1' = u_2$
$u_2' = u_3$
$u_3' = u_4$
$u_4' = \frac{d^4y}{dx^4} = -5\frac{d^2y}{dx^2} - \frac{x}{10}y^3 = -5u_3 - \frac{x}{10}u_1^3$

Let $\vec{F}(x_n, \vec{u}_n) \begin{pmatrix} u_2 \\ u_3 \\ u_4 \\ -5u_3 - \frac{x}{10}u_1^3 \end{pmatrix}$

Thought Process:

$\vec{k}_1 = \vec{F}(x_n, \vec{u}_1^n) \qquad \vec{u}_1^n \equiv \vec{u}_n$ at stage 1

$\vec{k}_2 = \vec{F}(x_n + \frac{h}{2}, \vec{u}_2^n)$ first component of $\vec{k}$

where $\vec{u}_2^n = \left(u_{1,1}^n + \frac{h}{2}k_{1,1}\ ,\ u_{1,2}^n + \frac{h}{2}k_{1,2}\ ,\ u_{1,3}^n + \frac{h}{2}k_{1,3}\ ,\ u_{1,4}^n + \frac{h}{2}k_{1,4}\right)$

$u_{1,1}^n \equiv$ first component of $\vec{u}_n$ at stage 1

$\vec{k}_3 = \vec{F}(x_n + \frac{h}{2}, \vec{u}_3^n)$

$\vec{u}_3^n = \left(u_{2,1}^n + \frac{h}{2}k_{2,1}\ ,\ u_{2,2}^n + \frac{h}{2}k_{2,2}\ ,\ u_{2,3}^n + \frac{h}{2}k_{2,3}\ ,\ u_{2,4}^n + \frac{h}{2}k_{2,4}\right)$

$\vec{k}_4 = \vec{F}(x_n + h, \vec{u}_4^n)$

$\vec{u}_4^n = \left(u_{3,1}^n + hk_{3,1}\ ,\ u_{3,2}^n + hk_{3,2}\ ,\ u_{3,3}^n + hk_{3,3}\ ,\ u_{3,4}^n + hk_{3,4}\right)$

Figure 4: Sketch of the reduction-to-first-order and higher-order Runge-Kutta workflow for Part 2.

B Data Files

- rk4_N_table.csv
- rk4_solution.csv
- pns3_solution.csv